GETTING STARTED WITH

Understanding Computer Networks: A Student-Friendly Conceptual Guide

Types of Networks, Topologies, Protocol Models, and Networking Devices Explained Clearly

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A Skilling Refcard provides concise reference guides on key technology topics for practitioners.

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REFCARD

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ABOUT COMPUTER NETWORKS

A computer network is a group of interconnected devices — such as computers, servers, smartphones, and IoT devices — that communicate and share resources through wired or wireless connections. Networking enables machines to exchange data efficiently, allowing people and applications to collaborate and access shared services.

The concept began in the late 1960s with ARPANET, the first packet-switching network, which laid the foundation for the modern internet. The development of TCP/IP protocols allowed different systems to communicate, while Ethernet standardized wired local networks. In the 1990s, the World Wide Web transformed the internet into a global platform for information, business, and entertainment.

Today, networks support almost every aspect of modern life, including messaging, streaming, online payments, cloud computing, healthcare systems, government services, and remote work. Key advantages of networking include resource sharing, centralized data storage, scalability, mobility, and improved efficiency.

A computer network consists of end devices, networking hardware (routers, switches), transmission media, communication protocols, and network software. Understanding how these components work together is essential for students, developers, and IT professionals managing modern digital infrastructure.

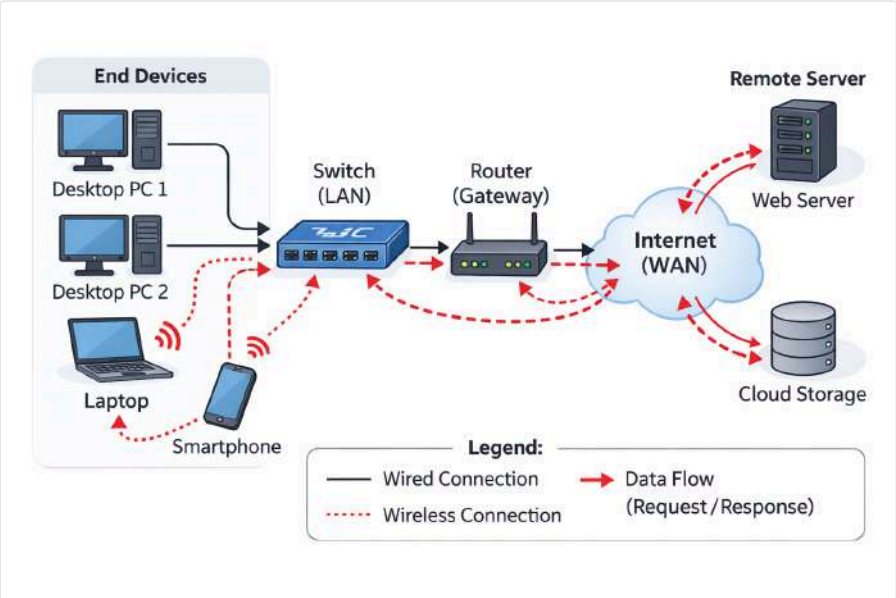


Figure 1: Basic Computer Network Overview

TYPES OF NETWORKS

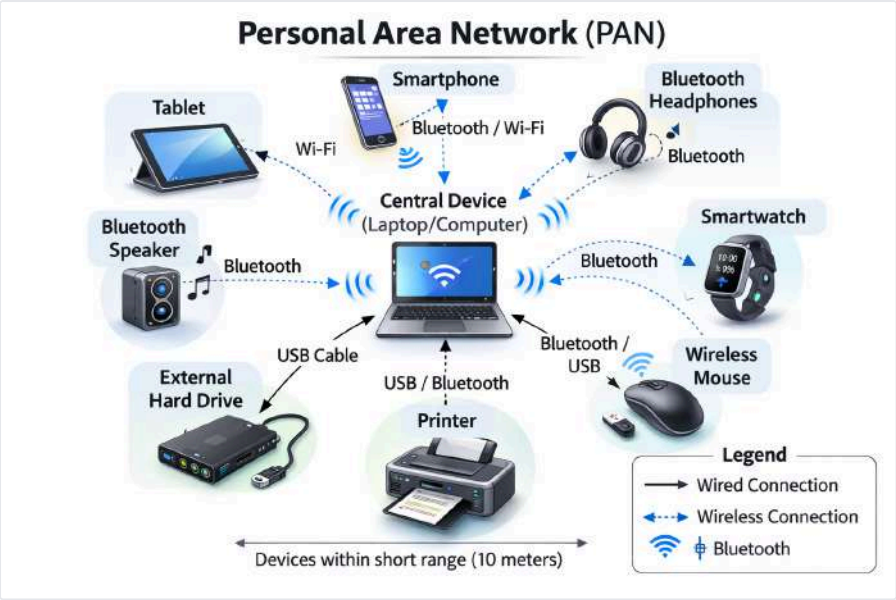
Computer networks are classified based on their geographical coverage, purpose, and scale. Understanding these categories helps you choose the right network design for any scenario — from connecting Bluetooth earbuds to your phone, to linking bank branches across an entire country.

Table 1: Types of Computer Networks

TYPE	FULL FORM	COVERAGE AREA	EXAMPLE
PAN	Personal Area Network	Few meters (personal space)	Bluetooth headphones connected to a phone
LAN	Local Area Network	Room, building, or small campus	Office Wi-Fi network with shared printers
MAN	Metropolitan Area Network	City or large campus	University campuses connected via fiber-optic backbone
WAN	Wide Area Network	Country, continent, or global	Bank branch connectivity across multiple cities
Internet	Global Network of Networks	Worldwide	Public internet accessed via ISPs

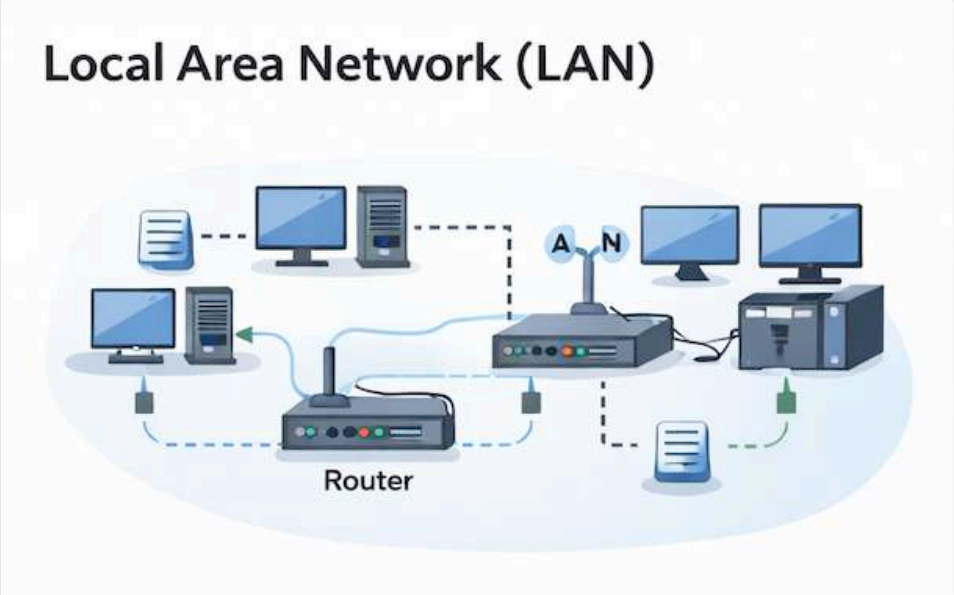
PERSONAL AREA NETWORK (PAN)

A Personal Area Network (PAN) is the smallest type of network and operates within a range of just a few meters. It is designed for personal use, typically connecting devices owned by a single individual. A common example is the Bluetooth connection between your smartphone and your wireless earbuds, or the link between a laptop and a wireless mouse. Another example is a USB tethering connection between a phone and a computer. PANs can be wired (using USB) or wireless (using Bluetooth or infrared). Because of their short range and limited number of devices, PANs are simple to set up and require minimal configuration. They are ideal for personal productivity and convenience.



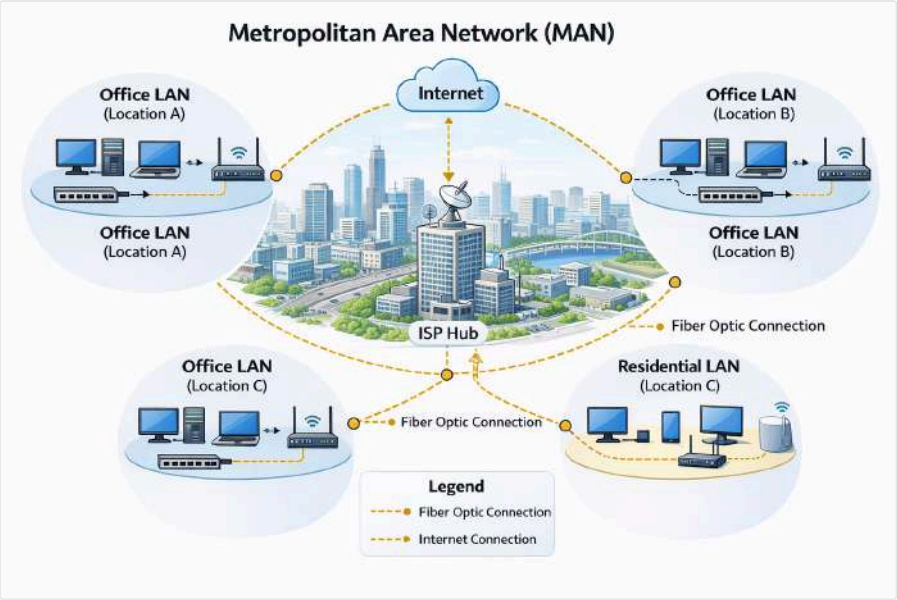
LOCAL AREA NETWORK (LAN)

A Local Area Network (LAN) covers a small geographic area, such as a single room, a floor, a building, or a small campus. LANs are the most common type of network you will encounter in offices, schools, and homes. When you connect your laptop to your home Wi-Fi router, you are part of a LAN. In an office, a LAN might connect dozens of computers, printers, and servers through Ethernet cables and switches. LANs offer high data transfer speeds — typically 100 Mbps to 10 Gbps — and are privately owned and managed. They allow devices to share resources like files, printers, and internet connections. Ethernet (IEEE 802.3) and Wi-Fi (IEEE 802.11) are the most common LAN technologies.



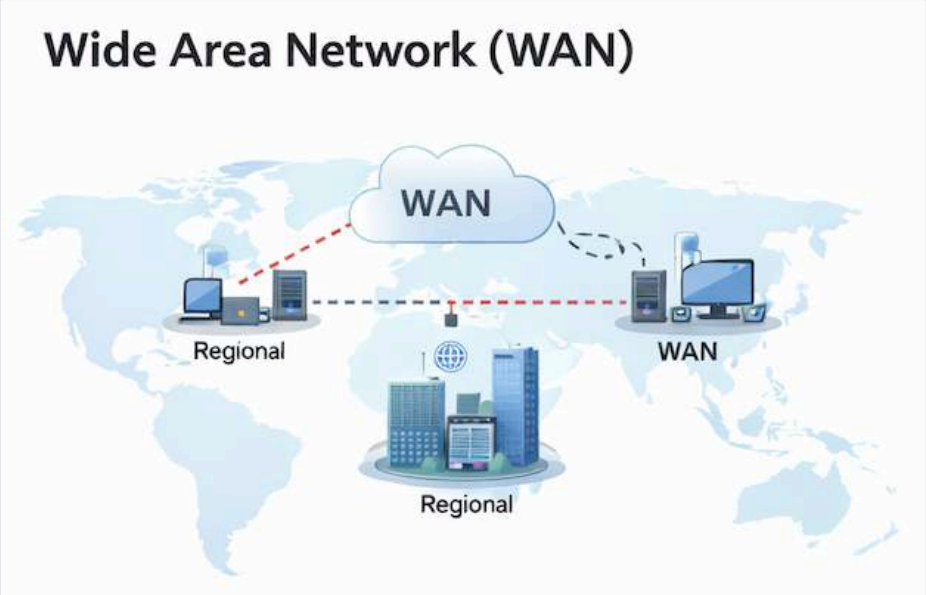
METROPOLITAN AREA NETWORK (MAN)

A Metropolitan Area Network (MAN) spans a larger geographic area than a LAN, typically covering a city or a large campus. Think of it as a network that connects multiple LANs spread across different buildings or locations within the same city. A university with multiple campuses connected through a fiber-optic backbone is a good example of a MAN. City-wide cable TV networks and municipal broadband services also fall into this category. MANs are usually owned and operated by large organizations, internet service providers (ISPs), or government entities. They use high-speed technologies like fiber optics and can cover distances of up to 50–100 kilometers.



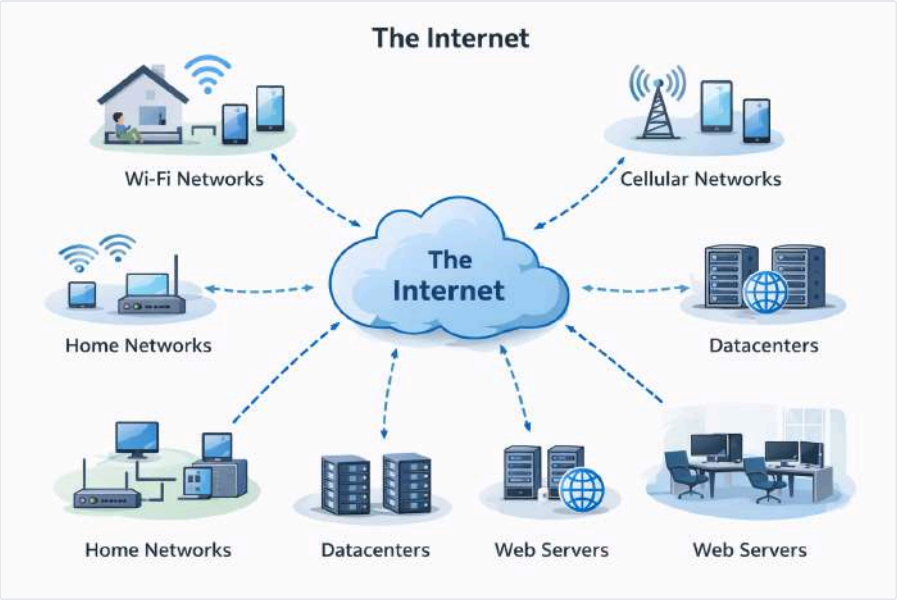
WIDE AREA NETWORK (WAN)

A Wide Area Network (WAN) covers a very large geographic area — a country, a continent, or even the entire world. WANs connect multiple LANs and MANs together using leased telecommunication lines, satellite links, or undersea fiber-optic cables. A classic example is a multinational bank that connects its branches in New York, London, and Mumbai through a private WAN so that they can share transaction data in real time. WANs typically have lower speeds than LANs due to the vast distances involved and the reliance on third-party infrastructure. The speeds depend on the service provider and the technology used — ranging from a few Mbps to several Gbps for premium connections.



INTERNET

The Internet is the largest and most well-known network in the world. It is essentially a global network of networks — a massive interconnection of millions of private, public, academic, business, and government networks linked by a broad array of technologies. The internet uses the TCP/IP protocol suite as its common communication language, allowing any connected device to communicate with any other connected device, regardless of location or manufacturer. Services like the World Wide Web, email, video conferencing, and online gaming all operate over the internet. What makes the internet unique is that it is not owned by any single entity — it is a collaborative, decentralized system governed by standards organizations like the IETF (Internet Engineering Task Force) and ICANN (Internet Corporation for Assigned Names and Numbers).



In summary, networks range from the tiny and personal (PAN) to the global and universal (Internet). Each type serves a distinct purpose and scale, and real-world IT infrastructure often involves multiple types working together — for example, your home LAN connects to a city MAN operated by your ISP, which in turn connects to a WAN that gives you access to the global internet.

NETWORK TOPOLOGIES

Network topology refers to the arrangement or layout of devices (called nodes) and the connections (called links) between them in a network. Topology defines how data flows through the network and has a direct impact on performance, reliability, scalability, and cost. There are two ways to think about topology: physical topology (how cables and devices are actually arranged) and logical topology (how data actually travels through the network, regardless of the physical layout). Here, we focus on the most common physical topologies.

In a Bus Topology, all devices are connected to a single central cable, called the bus or backbone. Data sent by any device travels in both directions along the bus, and every device on the network receives the data, but only the intended recipient processes it. Think of it like a single road with houses on both sides — a mail truck drives along the road and drops off letters, but each house only opens the letters addressed to it. Bus topology is simple and inexpensive to set up because it requires the least amount of cable. However, it has significant drawbacks. If the central cable fails, the entire network goes down. Performance degrades as more devices are added because all devices share the same communication channel, leading to collisions. For these reasons, bus topology is rarely used in modern networks, but it was common in early Ethernet networks (10BASE2 and 10BASE5).

In a Star Topology, every device is connected to a central device — usually a switch or a hub. All communication passes through this central device. If Device A wants to send data to Device B, the data goes from A to the central switch, and the switch forwards it to B. This is the most widely used topology in modern LANs. The main advantage of star topology is fault isolation: if one device or its cable fails, only that device is affected — the rest of the network continues to function normally. Star topology is also easy to add new devices to (just connect them to the switch) and easy to troubleshoot. The downside is that the central device is a single point of failure — if the switch goes down, the entire network stops working. Additionally, star topology requires more cable than bus topology since each device needs its own cable to the central device.

In a Ring Topology, devices are connected in a closed loop, where each device is connected to exactly two other devices — one on each side. Data travels in one direction around the ring (or in both directions in a dual-ring setup). Each device receives data from its upstream neighbor, checks if the data is addressed to itself, and if not, forwards it to the next device. Ring topology provides equal access to all devices and performs well under heavy load because data travels in an orderly, predictable manner. Token Ring networks used this topology, where a special token circulated the ring and a device could only transmit data when it held the token, preventing collisions. However, ring topology has a critical weakness: if any single device or connection in the ring fails, the entire network can be disrupted (unless a dual-ring configuration is used). Adding or removing devices also requires temporarily bringing down the network.

In a Mesh Topology, every device is connected to every other device in the network. This is called a full mesh. In a partial mesh, some devices are connected to all others, while some are connected only to the devices they exchange the most data with. Mesh topology offers the highest level of redundancy and reliability — if one connection fails, data can take an alternative path to reach its destination. This makes mesh ideal for critical applications where downtime is unacceptable, such as military communication networks or the backbone of the internet. The downside is cost and complexity: a full mesh with n devices requires n(n-1)/2 connections, which becomes extremely expensive as the number of devices grows. For example, a full mesh of 10 devices requires 45 connections. Configuration and maintenance are also more complex.

A Hybrid Topology is a combination of two or more different topologies. Most real-world enterprise networks are hybrid. For example, a large office building might use a star topology within each floor (devices connected to a floor switch), and the floor switches might be connected in a ring or mesh pattern to form the building's backbone. Hybrid topologies are flexible and scalable — they allow network designers to combine the strengths of different topologies while minimizing their weaknesses. The tradeoff is increased complexity in design, implementation, and troubleshooting.

A Tree Topology (also called a hierarchical topology) is a special type of hybrid that extends the star topology. It consists of a root node (usually a main switch or router) connected to multiple secondary nodes, which in turn connect to additional devices. It resembles an upside-down tree. This is common in large

organizations where the network is organized in layers — a core layer, a distribution layer, and an access layer. Tree topology is easy to manage and scale by adding branches, but like the star topology, failure of a node at a higher level can disconnect an entire branch.

Choosing the right topology depends on factors like budget, network size, reliability requirements, and future growth plans. In practice, understanding topologies helps you design networks that are efficient, resilient, and cost-effective.

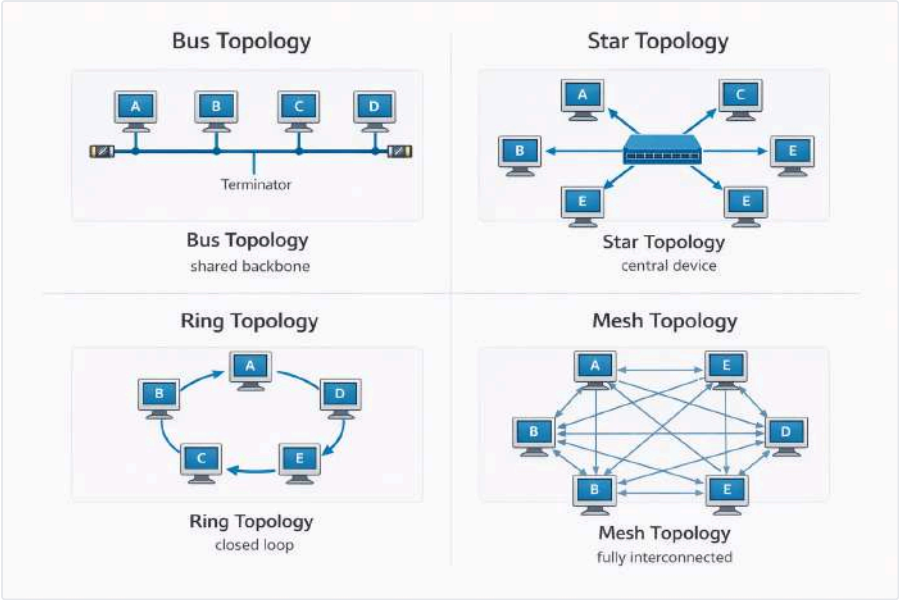


Figure 2: Common Network Topologies

OSI MODEL VS TCP/IP MODEL

🌐 NETWORKING COMMUNICATION MODELS — OSI AND TCP/IP

When two devices communicate over a network, a structured sequence of operations takes place. Data is generated by an application, formatted and segmented, logically addressed, routed across networks, transmitted as physical signals, and finally reconstructed and delivered to the correct application on the destination device. To standardize and organize this complexity, engineers use layered reference models — primarily the **OSI Model** and the **TCP/IP Model**.

🧩 THE OSI MODEL (7 LAYERS)

The **OSI (Open Systems Interconnection) Model**, developed by ISO in 1984, divides communication into seven logical layers. It is a conceptual and educational framework that provides a universal vocabulary for understanding and troubleshooting networking processes. While real-world systems do not implement it exactly as defined, it remains fundamental for structured networking knowledge.

Below, the OSI model is explained from **Layer 1 (bottom) to Layer 7 (top)**.

1. Physical Layer

The **Physical Layer** is responsible for transmitting raw binary data (bits) over a communication medium. It defines electrical signals, voltage levels, timing specifications, data rates, and physical components such as cables and connectors. It does not interpret data — it simply moves bits from one device to another.

Examples:

- `Cat5`, `Cat6`
- Fiber optic cables
- `RJ-45` connectors

Devices:

- Hubs
- Repeaters

Scenario:

In a real-world scenario, when a computer sends data through an Ethernet cable, the Physical Layer converts binary data into electrical pulses (or light pulses in fiber optics) and transmits them across the medium to the receiving device.

This layer does not understand frames or packets — only raw bits.

2. Data Link Layer

The **Data Link Layer** ensures reliable communication between two directly connected devices on the same local network. It packages raw bits into frames, adds MAC addresses for local identification, and performs error detection using mechanisms such as CRC. This layer ensures that frames are delivered correctly within the same network segment.

Sub-layers:

- Logical Link Control (LLC)

- MAC

Devices:

- Switches
- Bridges

Scenario:

For example, when your laptop sends data to your local router, the Data Link Layer adds the router's MAC address to the frame. A network switch reads this MAC address and forwards the frame to the correct device within the LAN.

3. Network Layer

The **Network Layer** handles logical addressing and routing between different networks. It assigns IP addresses and determines the best path for packets to travel across interconnected networks. Routers operate at this layer and forward packets based on routing tables.

Protocols:

- `IP`
- `ICMP`
- `ARP`

Devices:

- Routers

Scenario:

In a practical scenario, when you access a website hosted in another country, the Network Layer ensures that packets travel across multiple routers and networks using IP addresses until they reach the destination server.

This layer enables internetwork communication.

4. Transport Layer

The **Transport Layer** provides end-to-end communication between applications running on different devices. It breaks data into segments, assigns port numbers to identify specific applications, manages flow control, detects errors, and may retransmit lost data. The two primary protocols at this layer are TCP (reliable and connection-oriented) and UDP (fast and connectionless).

Protocols:

- `TCP` (reliable, connection-oriented)
- `UDP` (fast, connectionless)

Scenario:

For instance, when loading a webpage, TCP establishes a connection using a three-way handshake, ensures that data arrives in order, and retransmits any lost segments to maintain reliability.

This layer ensures reliability when required.

5. Session Layer

The **Session Layer** manages communication sessions between two systems. It establishes, maintains, synchronizes, and terminates sessions. It also supports checkpoints to allow recovery if communication is interrupted.

Scenario:

For example, during a large file transfer, the Session Layer may create synchronization checkpoints. If the connection drops halfway through, the transfer can resume from the last checkpoint instead of starting from the beginning.

It controls dialog between systems.

6. Presentation Layer

The **Presentation Layer** ensures that data sent by one system can be properly understood by another. It handles data formatting, character encoding, encryption, and compression. This layer acts as a translator between different data representation formats.

Examples:

- `SSL/TLS` encryption
- ASCII ↔ EBCDIC conversion
- Image compression

Scenario:

In a secure web session, the Presentation Layer encrypts data using TLS before transmission. On the receiving side, it decrypts the data so the application can interpret it correctly.

This layer focuses on syntax and representation.

7. Application Layer

The **Application Layer** is the layer closest to the end user. It provides network services to user applications and enables interaction with network protocols such as HTTP, FTP, SMTP, and DNS. This layer allows applications to request and receive network resources.

Protocols:

- `HTTP`
- `SMTP`
- `FTP`
- `DNS`

Scenario:

When you type a URL into a browser, the Application Layer generates an HTTP request and initiates communication with the web server.

This is the layer closest to the user.

🔄 COMPLETE OSI COMMUNICATION FLOW (EXAMPLE)

Example: Opening a website.

1. Application Layer → Browser creates `HTTP` request
2. Presentation Layer → Data encrypted via `TLS`
3. Session Layer → Session established
4. Transport Layer → `TCP` handshake and segmentation
5. Network Layer → IP routing across networks
6. Data Link Layer → Frame delivered within local network
7. Physical Layer → Bits transmitted on the wire

At the receiving device:

- Data moves upward from Layer 1 → Layer 7
- Each layer removes its corresponding header
- The web page is delivered to the browser

🌍 THE TCP/IP MODEL (4 LAYERS)

The **TCP/IP Model**, developed by the U.S. Department of Defense, is the practical foundation of the modern internet.

Key characteristics:

- Directly implemented in real-world systems
- Protocol-driven architecture
- Simpler than OSI

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1. Network Access Layer

Combines OSI Layers 1 and 2.

The **Network Access Layer** combines the responsibilities of the OSI Physical and Data Link layers. It manages framing, MAC addressing, and physical transmission over technologies such as Ethernet and Wi-Fi.

Technologies:

- Ethernet
- Wi-Fi
- PPP
- DSL

Scenario:

In practice, when your computer connects to a Wi-Fi router, this layer handles both framing and the wireless signal transmission.

2. Internet Layer

The **Internet Layer** corresponds to the OSI Network Layer. It manages logical addressing and routing using IP. Protocols such as IP, ICMP, and ARP operate here.

Protocols:

- `IP` (IPv4, IPv6)
- `ICMP`
- `ARP`

Scenario:

For example, when sending data to a remote server, this layer determines how packets travel across interconnected networks.

3. Transport Layer

The **Transport Layer** provides end-to-end communication services using TCP or UDP. TCP ensures reliable delivery with acknowledgments and retransmissions, while UDP provides faster, connectionless transmission.

Protocols:

- `TCP`
- `UDP`

Scenario:

Streaming applications often use UDP for speed, whereas web browsing typically uses TCP for reliability.

Provides end-to-end communication services.

4. Application Layer

The **Application Layer** in TCP/IP combines the functions of the OSI Session, Presentation, and Application layers. It handles application logic, formatting, encryption (when implemented within protocols), and user interaction.

Protocols:

- `HTTP`
- `FTP`
- `SMTP`
- `DNS`
- `SSH`

Scenario:

When accessing a website, protocols such as HTTP and HTTPS operate at this layer.

Handles application logic, formatting, and session management.

🎯 KEY DIFFERENCE

The OSI model is an idealized conceptual framework designed for understanding and teaching networking principles. The TCP/IP model is a practical implementation model that reflects how the internet actually operates. In real-world environments, networking follows TCP/IP protocols, but troubleshooting explanations often reference OSI layers for clarity.

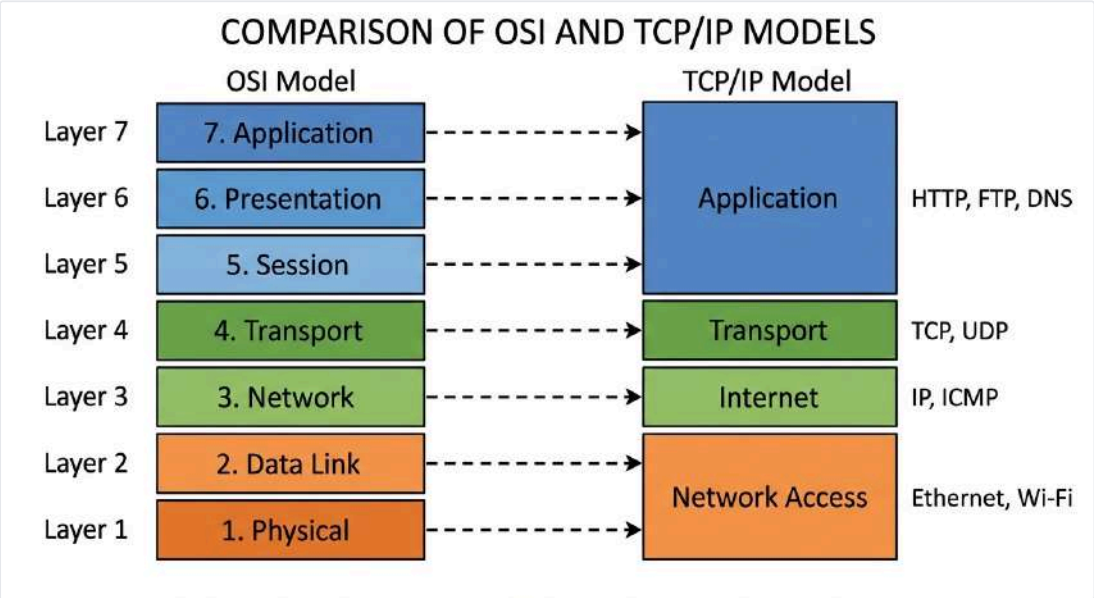

Figure 3: OSI vs TCP/IP Layer Mapping

Table 2: OSI and TCP/IP Layer Comparison

OSI LAYER	LAYER NUMBER	TCP/IP EQUIVALENT	MAIN FUNCTION	EXAMPLE PROTOCOLS/DEVICES
Application	7	Application	Provides network services to end-user applications	HTTP, FTP, SMTP, DNS
Presentation	6	Application	Data translation, encryption, and compression	SSL/TLS, JPEG, ASCII
Session	5	Application	Establishes, manages, and terminates sessions	NetBIOS, PPTP, RPC
Transport	4	Transport	Reliable end-to-end data delivery and flow control	TCP, UDP
Network	3	Internet	Logical addressing and routing across networks	IP, ICMP, Routers
Data Link	2	Network Access	MAC addressing, framing, and error detection	Ethernet, Wi-Fi, Switches
Physical	1	Network Access	Transmission of raw bits over physical media	Cables, Hubs, Signals

TCP VS UDP

TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are the two main protocols that operate at the Transport Layer (Layer 4) of the OSI model and the Transport Layer of the TCP/IP model. Every application that communicates over a network uses either TCP or UDP (or in rare cases, both) to send and receive data. Understanding the differences between them is essential for networking and software development.

TCP is a connection-oriented protocol. This means that before any data is exchanged, the sender and receiver must first establish a connection through a process called the three-way handshake. In this handshake, the sender sends a SYN (synchronize) packet to the receiver. The receiver responds with a SYN-ACK (synchronize-acknowledge) packet. The sender then replies with an ACK (acknowledge) packet. Only after this three-step process is complete can data transmission begin. This connection setup ensures that both parties are ready and able to communicate.

TCP provides reliable data delivery. Every segment of data sent using TCP is assigned a sequence number, and the receiver must send an acknowledgment (ACK) for each segment it receives. If the sender does not receive an acknowledgment within a certain time, it automatically retransmits the missing segment. This mechanism ensures that no data is lost, duplicated, or delivered out of order. TCP also provides flow control — the receiver can tell the sender to slow down if it is being overwhelmed with data — and congestion control — TCP reduces its sending rate when it detects network congestion. These features make TCP highly reliable but add overhead, making it slower than UDP.

TCP is used by applications where data integrity is more important than speed. Web browsing (HTTP/HTTPS) relies on TCP because every part of a web page — HTML, CSS, images — must arrive correctly and in the right order for the page to render properly. Email (SMTP, IMAP, POP3) uses TCP because losing even a small part of a message would make it unreadable. File transfer (FTP, SFTP) uses TCP because a corrupted file is useless. Database queries, API calls, and SSH remote sessions also use TCP for the same reason.

UDP is a connectionless protocol. There is no handshake, no connection setup, and no connection teardown. The sender simply packages the data into a datagram and sends it to the receiver. There is no guarantee that the datagram will arrive, no guarantee that it will arrive in order, and no mechanism for the receiver to request retransmission of lost data. UDP includes a basic checksum for error detection, but if a datagram is corrupted, it is simply discarded — there is no retry.

This lack of overhead is what makes UDP fast. Without the need for connection setup, acknowledgments, sequencing, and retransmission, UDP can send data with minimal latency. This makes it ideal for applications where speed is critical and occasional data loss is acceptable.

Video streaming is a classic UDP use case. When you watch a live stream on Twitch or a video on YouTube (which uses adaptive streaming over HTTP but often leverages UDP-based protocols like QUIC internally), losing a single video frame is barely noticeable — the stream simply continues. Retransmitting that lost frame would cause a delay that is far more disruptive than the minor quality glitch from the lost data. Online gaming is another major UDP use case. In a fast-paced multiplayer game, the positions of players, projectiles, and objects must be updated many times per second. If one update is lost, the next update will contain the latest state anyway, making retransmission unnecessary and counterproductive. VoIP (Voice over IP) services like phone calls over the internet also use UDP — a tiny gap in audio is preferable to the delay caused by retransmitting lost packets.

DNS (Domain Name System) queries also use UDP by default. When your browser needs to resolve a domain name like www.example.com to an IP address, it sends a small UDP query to a DNS server and expects a small response. The entire exchange fits in a single datagram in each direction, so the overhead of setting up a TCP connection would be wasteful. However, if the DNS response is too large to fit in a single UDP datagram (more than 512 bytes traditionally, or 4096 bytes with EDNS), DNS falls back to TCP.

A more recent development is the QUIC protocol, developed by Google and now used by HTTP/3. QUIC is built on top of UDP but adds its own connection management, encryption (built-in TLS 1.3), and reliability features. It demonstrates that the line between TCP and UDP is becoming more nuanced — developers can build reliability on top of UDP when they need fine-grained control over the transport behavior.

In summary, TCP and UDP are not competitors — they are tools designed for different jobs. TCP is your choice when you need every byte to arrive correctly and in order. UDP is your choice when you need speed and can tolerate some data loss. A well-designed application chooses the right protocol based on its specific requirements.

Table 3: TCP vs UDP Comparison

FEATURE	TCP	UDP
Connection Type	Connection-oriented (three-way handshake)	Connectionless (no handshake)
Reliability	Reliable — acknowledgments, retransmission, sequencing	Unreliable — no acknowledgments or retransmission
Ordering	Guarantees in-order delivery using sequence numbers	No ordering guarantee — datagrams may arrive out of order
Speed	Slower due to overhead (ACKs, flow control, retransmission)	Faster due to minimal overhead
Error Checking	Comprehensive — checksum, ACK, retransmission	Basic checksum only — corrupted packets are discarded
Flow Control	Yes — receiver can regulate sender's data rate	No — sender transmits at its own rate
Congestion Control	Yes — TCP reduces rate during network congestion	No — UDP does not adapt to congestion
Header Size	20–60 bytes	8 bytes
Data Unit	Segment	Datagram
Use Cases	Web browsing (HTTP), Email (SMTP), File transfer (FTP), SSH	Video streaming, Online gaming, VoIP, DNS queries

HUB VS SWITCH VS ROUTER

Hubs, switches, and routers are the three fundamental networking devices that students encounter when studying computer networks. While they may look similar physically — boxes with multiple Ethernet ports — they operate at different layers of the OSI model and serve very different purposes. Understanding the differences between them is critical for designing efficient networks and troubleshooting connectivity problems.

A Hub is the simplest and most basic networking device. It operates at Layer 1 (Physical Layer) of the OSI model. A hub is essentially a multiport repeater — when it receives a data signal on one port, it blindly replicates and broadcasts that signal out of every other port. It does not look at the data, does not understand addresses, and does not make any intelligent forwarding decisions. Imagine a person in a room shouting a message — everyone in the room hears it, but only the intended recipient pays attention to it. That is exactly how a hub works.

Because a hub broadcasts every frame to every port, it creates a single collision domain — all devices connected to the hub share the same bandwidth and must take turns transmitting. If two devices send data at the same time, a collision occurs, and both must retransmit. This dramatically reduces performance as more devices are added. Hubs were common in early Ethernet networks (10BASE-T) but have been almost entirely replaced by switches in modern networks. You may still encounter hubs in very old installations, exam questions, or low-cost hobby setups.

A Switch is a significantly more intelligent device. It operates at Layer 2 (Data Link Layer) of the OSI model and makes forwarding decisions based on MAC (Media Access Control) addresses. When a switch receives a frame on one of its ports, it reads the destination MAC address in the frame header and forwards the frame only to the specific port where the destination device is connected. It does not broadcast the frame to all ports (except in special cases like broadcast frames or when it does not yet know which port a device is on).

The switch learns which devices are connected to which ports by building and maintaining a MAC address table (also called a CAM table — Content Addressable Memory table). When a frame arrives, the switch records the source MAC address and the port it arrived on. Over time, the switch builds a complete map of which MAC addresses are reachable through which ports. This intelligent forwarding creates separate collision domains for each port, meaning devices can transmit simultaneously without collisions. This is why switches provide much better performance than hubs.

Switches also support full-duplex communication, meaning a device can send and receive data at the same time — unlike hubs, which only support half-duplex. Modern managed switches offer additional features like VLANs (Virtual LANs), which allow you to logically segment a single physical network into multiple isolated networks, improving security and performance. Other features include Quality of Service (QoS) for prioritizing traffic, port mirroring for monitoring, and Spanning Tree Protocol (STP) for preventing loops.

A Router is the most sophisticated of the three devices and operates at Layer 3 (Network Layer) of the OSI model. While a switch forwards data within a single network based on MAC addresses, a router forwards data between different networks based on IP addresses. The router examines the destination IP address in each packet, consults its routing table (a database of known networks and the best paths to reach them), and forwards the packet to the appropriate next hop — which might be another router, a switch, or the destination device itself.

Routers are what make the internet possible. When you send a request from your home computer to a server on the other side of the world, that request passes through dozens of routers, each one making an independent forwarding decision to move the packet closer to its destination. Your home router is the gateway between your local network (LAN) and the internet (WAN). It typically performs several additional functions: Network Address Translation (NAT), which allows multiple devices on your private network to share a single public IP address; DHCP (Dynamic Host Configuration Protocol), which automatically assigns IP addresses to devices on your network; and firewall functionality, which filters incoming and outgoing traffic based on security rules.

Routers also separate broadcast domains. In a network connected by switches alone, a broadcast frame sent by one device reaches every other device on the network. A router stops broadcasts from propagating from one network to another, which is important for performance and security in large networks.

To use a real-world analogy: a hub is like a loudspeaker that blasts every message to everyone in the building. A switch is like a mail clerk who reads the name on each envelope and delivers it to the correct desk. A router is like a postal service that reads the city and zip code on each letter and routes it to the correct post office in the correct city.

In modern home and small office networks, these functions are often combined into a single device. Your home Wi-Fi router is actually a router, a switch (the Ethernet ports on the back), a wireless access point (the Wi-Fi radio), and often a modem — all in one box. However, in enterprise and data center environments,

these devices are separate, specialized, and much more powerful.

Understanding the distinction between hubs, switches, and routers is foundational for network design, troubleshooting, and understanding how data travels across local and global networks.

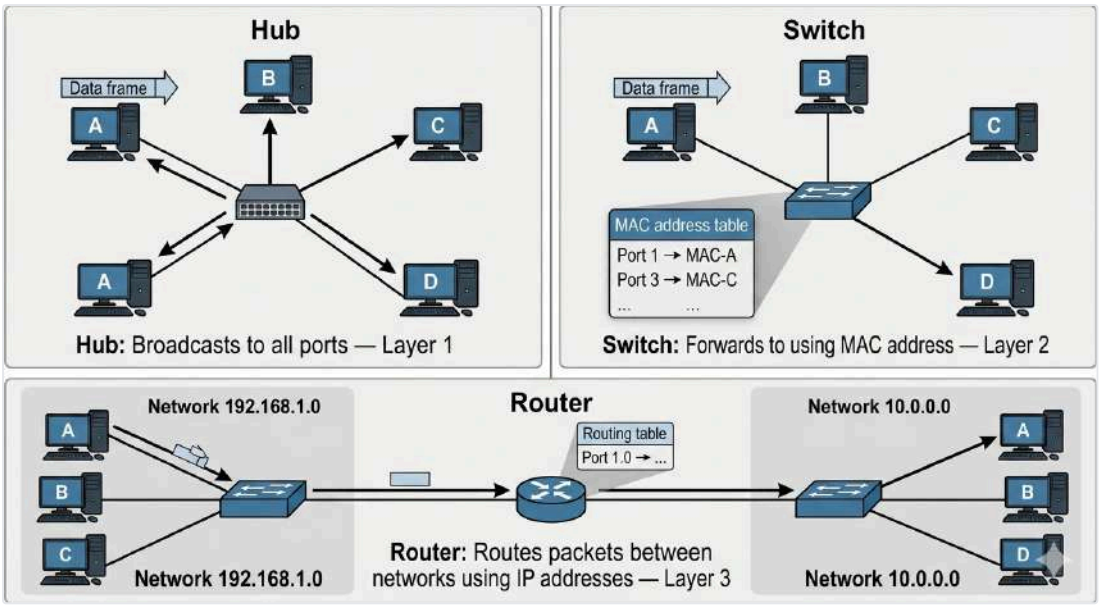

Figure 4: Hub vs Switch vs Router Functionality

Table 4: Networking Device Comparison

FEATURE	HUB	SWITCH	ROUTER
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Used	None	MAC Address	IP Address
Main Function	Broadcasts data to all connected ports	Forwards frames to specific port based on MAC table	Routes packets between different networks using routing table
Intelligence	None — simple signal repeater	Medium — learns MAC addresses	High — makes routing decisions, supports NAT, DHCP, firewall
Collision Domain	Single shared collision domain	Separate collision domain per port	Separate collision domain per port
Broadcast Domain	Single broadcast domain	Single broadcast domain (unless VLANs configured)	Separates broadcast domains
Speed Impact	Degrades with more devices	Maintains performance with more devices	Depends on routing complexity and throughput
Duplex	Half-duplex only	Full-duplex supported	Full-duplex supported
Cost	Very low (obsolete)	Moderate	Higher
Modern Usage	Virtually obsolete	Core device in all LANs	Gateway between LAN and WAN, backbone of the internet

CONCLUSION

This Refcard has covered the essential building blocks of computer networking — the concepts that every student, developer, and IT professional must understand before diving into more advanced topics like network security, cloud computing, or distributed systems.

We began with the fundamentals: a computer network is a system of interconnected devices that share resources and communicate using agreed-upon rules called protocols. Networks have evolved from small experimental systems like ARPANET into the massive global infrastructure that powers virtually every digital service we use today.

We explored the types of networks based on their geographic scope. PANs cover personal devices within arm's reach. LANs connect devices within a home, office, or building. MANs span cities and large campuses. WANs connect locations across countries and continents. The internet ties all of these together into a single global network of networks.

We examined network topologies — the physical and logical arrangements of devices and connections. Bus topology uses a shared backbone cable. Star topology connects all devices to a central switch. Ring topology forms a closed loop. Mesh topology provides maximum redundancy through direct interconnections. Hybrid and tree topologies combine these patterns to meet real-world needs. Each topology offers different tradeoffs in cost, reliability, and scalability.

We compared the OSI and TCP/IP models — the two layered frameworks for understanding network communication. The OSI model's seven layers provide a detailed theoretical framework for learning. The TCP/IP model's four layers reflect the practical architecture of the internet. Understanding both models and how they map to each other is essential for exams, interviews, and real-world troubleshooting.

We contrasted TCP and UDP — the two transport layer protocols. TCP provides reliable, ordered, connection-oriented delivery for applications like web browsing, email, and file transfer. UDP provides fast, lightweight, connectionless delivery for applications like video streaming, online gaming, and DNS queries. The choice between them depends on whether your application prioritizes reliability or speed.

Finally, we compared hubs, switches, and routers — the three foundational networking devices. Hubs are simple Layer 1 broadcasters that are now obsolete. Switches are intelligent Layer 2 devices that forward frames based on MAC addresses and form the core of every modern LAN. Routers are sophisticated Layer 3 devices that route packets between different networks using IP addresses and are the backbone of the internet.

Quick Revision Checklist: (1) Networks enable resource sharing, communication, and centralized management. (2) PAN < LAN < MAN < WAN < Internet in terms of coverage. (3) Star topology is the most common in modern LANs. (4) OSI has 7 layers (theoretical); TCP/IP has 4 layers (practical). (5) TCP is reliable and slow; UDP is fast and unreliable. (6) Hubs broadcast, switches forward intelligently, routers route between networks. (7) Know which OSI layer each device and protocol operates at. Master these fundamentals, and you will have a solid foundation for everything else in networking.